

4. The Government intends to phase out the use of petrol and diesel cars by 2040 and switch to electric vehicles. Two such electric vehicles are the Voltsa and the Ampra. The Voltsa has an electric motor only. The Ampra has both an electric motor and a petrol engine. The table gives information about these electric vehicles and a petrol only car.

	Voltsa	Ampra	Petrol only
Distance travelled on one tank of fuel (km)		450	600
Distance travelled on one charge (km)	150	60	
Cost of one full charge (p)	264	90	
Mean volume of petrol used (litres/100 km)	0	4	7
Cost of petrol (p/litre)		120	120
Mean CO ₂ produced over 100 km (g/100 km)	0	7 000	12 000

Use the information in the table to answer the following questions.

- (a) (i) Eric travels 110 km each day. He is concerned about his carbon footprint. Why would he choose a Voltsa rather than an Ampra? [1]

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- (ii) Calculate the mass of CO₂ produced if the Ampra is driven 250 km. [2]

Mass of CO₂ = g

- (b) (i) The Voltsa can be charged fully in six hours using a 4 kW charger. When 1 kWh of electricity is generated it produces 0.4 kg of carbon dioxide. Calculate how much CO₂ is produced to fully charge the Voltsa. [2]

Mass of CO₂ = kg

- (ii) Ian says that the CO₂ data for the Voltsa is misleading. Explain why. [2]

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- (c) Ian drives 600 km every week. He says he is not going to buy the more expensive **Volt**sa since it will not save him money to run compared to his **petrol only** car. Explain whether you agree with Ian. [2]

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